

Project lifting management plan

Table of contents

1. Objectif.....	2
2. Scope.....	2
3. Documents de références	2
4. Definitions et abreviations.....	3
5. Roles and Responsibilities.....	4
5.1. Project manager/project director	4
5.2. Lifting designated person	4
5.3. Superviseur levage.....	4
5.4. Signaller	5
5.5. Lifting operator.....	5
5.6. Slinger	6
5.7. H&S manager/H&S supervisor	6
5.8. All personnel involved	6
6. Training, competencies and authorizations.....	6
6.1. Training.....	6
6.2. Lifting team members competencies	7
6.3. Authorization to operate.....	8
7. Equipment conformity.....	8
8. Lifting management documentation on projects	9
8.1. Lifting equipment utilization plan	9
8.2. Lifting operation plan	10
9. Preparation and execution of lifting operations.....	12
9.1. Installation of lifting equipment.....	12
9.2. Review before start	12
9.3. Lifting execution	12
10. Loan of lifting equipment.....	13

1. Objectif

The objective of this document is to describe the general measures to be taken by the Project/site to ensure lifting operations are carried out safely.

It depicts, amongst other subjects, the roles and responsibilities of the various personnel involved in lifting operations, as well as the minimum requirements to provide conformance with prevailing regulations, codes and standards governing lifting operations on the project/site.

2. Scope

This document applies to all lifting operations carried out on the project using mechanically operated lifting equipment.

This concerns, among others, the following equipment:

- mobile telescopic boom cranes
- mobile and crawler lattice boom cranes
- lorry cranes
- tower cranes
- excavators
- launching girders and lifting frames
- telehandlers
- winches and chain hoists (for a load >100kg)
- overhead cranes - Gantry cranes
- lifting with helicopter

This document applies to all companies working on sites (including service providers and subcontractors).

This document does not apply to lifting of personnel or to manual lifting with ropes.

3. Documents de références

BYCN – Lifting Directive

BYTP-REF-H&S-2210 – Standard Lifting

ED 6178 – INRS France - Lifting accessories - Slinger's handbook - September (2014)

ED 6107 - INRS France - Mobile crane - Safety manual (2018)

ED 0813 - INRS France - Tower crane - Safety manual (April 2009)

Good practices guide - Union Française du Levage <French Lifting Union> (Nov 2018)

Guide to application of the Machinery Directive EN 206/48/EC (June 2010)

4. Definitions et abreviations

Topic / Abbreviation	Description
Lifting operation	Any operation concerned with the lifting or lowering of loads using lifting equipment.
Lifting equipment	Any equipment or machinery used for lifting or lowering loads, including attachments and accessories used for anchoring, fixing or supporting the equipment. This includes, but is not limited to cranes, lifting jacks, excavators equipped for lifting and forklift. Lift tables and pallet trolleys are not considered to be lifting equipment.
Lifting accessories	Components not permanently attached to the lifting equipment, which are placed between the equipment and the load, or attached to the load itself. They include, among others, hooks, shackles, lifting/spreader beams, lifting slings, mechanical fittings, shim plates...etc.
Suitability examination	Thorough examination or review by a suitably appointed competent person to ensure compatibility between a lifting equipment and its intended use, taking into account the environment in which the equipment will be used and technical data provided by its manufacturer.
Lifting Plan	Gathers for a specific lift according to BYCN Lifting Directive: • Lifting Equipment Utilisation Plan • Lifting Operation Plan
Project lifting management plan	This document outlines, for a project/site, the organizational, technical and resource measures applicable for all lifting activities conducted at site/project
Lifting Equipment Utilisation Plan	This document sets out, for a given lifting equipment, all the conditions under which the equipment will be used on the project/site: Suitability examination of the lifting equipment, coordination between the various people involved, risk analysis...
Lifting Operation Plan (for a given lifting equipment and a type of load to be lifted)	This document sets out the technical means to be used for lifting a type of load, including methods of slinging, as well as organizational measures to be complied with by the various personnel involved in the lifting operation
Autorisation to Operate	Document attesting the recognition by the employer, or his representative, of the ability of one of his employees to safely operate the construction plant mentioned on the authorization
Periodic General Inspection	The purpose of periodic general inspections is to assess the condition of the equipment's components and safety systems for which any deterioration could result in dangerous situations in order to determine: • - if a repair or a replacement is necessary without undue delay. • - if the safety devices/systems can fulfil their designated function properly.

These regular inspections are not merely to check the overall working order of an item of equipment but are a careful examination of its components and safety systems.

5. Roles and Responsibilities

Appendix **Roles of Lifting Personnel [BYTP-INF-H&S-2135]** outlines the qualifications, duties and know-how of each personnel involved in a lifting operation

5.1. Project manager/project director

identifies and assigns formally each of the following roles within the organization chart of the project/site or of the entity to which the project/site depends:

- Lifting Designated Person (PM/PD can play this role when the managerial organization of the project/site is limited)
- Lifting Supervisor(s)
- ensures that the other lifting personnel are identified, certified and receive their authorizations to operate according to their training and level of expertise
- ensures that the *Project lifting management plan* is established, implemented and communicated to Subcontractors and service providers in order for them to organize accordingly their own lifting operations
- audits the implementation of the *Project lifting management plan* and the compliance with the BYTP lifting requirements
- defines the means necessary to carry out verification of equipment at site, including the Periodic General Inspection and ensure that these verifications are conducted in due time

5.2. Lifting designated person

- responsible for the implementation of the lifting standards within their perimeter
- issues or validates the *Project lifting management plan*
- supports the Project Manager/Director in assigning the lifting Supervisor(s)
- supports the project/site teams in preparing the *Project lifting management plan*, the *Lifting Equipment Utilization Plan* and the *Lifting Operation Plan*, when necessary
- Participates to audits of the implementation of the *Project lifting management plan* and the compliancy with the BYTP lifting requirements
- Participates in risk analysis of lifting operations planned on his project/site

5.3. Superviseur levage

- responsible for the implementation, by the teams he supervises, of the *Lifting Equipment Utilization Plans* and the *Lifting Operation Plans*
- participates in the development of the *Lifting Equipment Utilization Plans*, validates them and ensures they are maintained up-to-date
- issues the "specific" and "complex" *Lifting Operation Plans* and communicate them to the Signallers
- coordinates the Signallers under his responsibility
- checks lifting equipment and materials to ensure they are compliant on their reception at site, have the required certificates and are maintained and inspected

- ensures that lifting equipment and accessories are used within the strict limits of their suitability examination (initial and subsequent, if conditions of use change), from the moment the equipment/accessories are delivered and accepted on site until their demobilization, dismantling or removal
- ensures that lifting equipment is assembled and dismantled in strict accordance with manufacturer's instruction
- takes direct control of "specific" and "complex" lifting operations and issue a Lifting Permit for each "complex" operation

5.4. Signaller

- Must be present for all lifting, slewing and lowering of loads. He is attired in such a way (preferably a specific high-visibility vest) that he can be distinctively identified by lifting operators
- guides the load (without touching the load)
- appoints and supervises the slingers involved in his lifting operations
- verifies that the type of lifting has been correctly evaluated in the *Lifting Operation Plan* and that the plan has been correctly filled-in
- checks before authorizing any lift: condition of the lifting equipment and load, slings, weather conditions, working environment, team involved, communication means, adequacy with the *Lifting Operation Plan*, validity of the *Lifting Equipment Utilization Plan*, etc.
- conducts a briefing of the team involved in the lifting before the operation starts and informs other parties working in the area
- directly and continuously supervises the lifting operations and gives instructions to the various involved persons (lifting operators and slingers)
- does not start (or stops) his lifting operation in the event of deviation from the established *Lifting Operation Plan* and/or in case of immediate hazard, including in case of unforeseen co-activity or presence of personnel not involved in the lifting operation
- gives authorization to slacken and detach slings at the end of the lift, after having ensured the stability of the load

5.5. Lifting operator

- operates the lifting equipment, such as a tower/mobile crane, gantry crane etc. and is responsible for operating the lifting equipment within its operating limits and in accordance with all guidance and rules
- operates and maintain his lifting equipment according to the manufacturer's requirements and instructions
- inspects the condition and checks the functioning of the lifting equipment/safety devices prior to use in accordance with the manufacturer's instructions and site rules
- operates the lifting equipment according to the *Lifting Equipment Utilization Plans* and the *Lifting Operation Plans* and follows the instructions given by the Signaller. However, he is required to stop if instructions are unclear, if he is or becomes aware of a hazard or if he is unsure about conditions

5.6. Slinger

- Strictly respects the slinging and unslinging rules
- checks lifting accessories to ensure they are safe and suitable for the intended purpose as set out in the *Lifting Operation Plan*
- guides the load during lifting or lowering (and slewing if load at a low level) using a device that can be used safely to prevent the load from striking anything or getting stuck and to allow the load to be steered in all directions (taglines, push sticks, etc.). Does not guide the load by directly putting hands on it
- Keeps always within sight and view of the signaller and as far as practical the crane operator
- stay out from under a suspended load

5.7. H&S manager/H&S supervisor

- supports, when necessary, the Project Director/Manager for the implementation of this *Lifting Operation Standard*
- participates to audits of the implementation of the *Project lifting management plan* and to the compliance with lifting standards
- participate in risk analysis of lifting operations planned on his project/site
- alerts and stops any lifting operation when significant or repeated deviations are observed (by CWD, subcontractor or service provider) and reports to the Project Director/Manager to ensure that correctives actions are conducted

5.8. All personnel involved

Any employee involved in lifting operations:

- Always stands in a safe place during lifting and never pass below or place any part of his body under a suspended load or between a suspended load and a fixed object
- ensures no other person is located under a suspended load or between a suspended load and a fixed object
- never climbs on a load during its lifting
- pays attention to suspended loads, signals from signallers and to any lifting equipment in operation
- Alerts and stops any lifting operation when significant or repeated deviations are observed (by CWD, subcontractor or service provider) and reports to the Project Director/Manager to ensure that correctives actions are conducted

6. Training, competencies and authorizations

6.1. Training

Any person involved in lifting operations (Lifting Designated Person, Supervisor, Signaller, Slinger, Lifting Operator, etc.) is a person who is competent in his field:

- trained (with regular refresher training)
- qualified or certified as necessary for the position
- authorized by the Project Director/Manager or the Lifting Designated Person, as per his training or level of expertise

A training program is defined and implemented by the project/site for the various personnel involved in lifting operations. These training courses can be delivered internally by competent

CWD personnel or by competent external training providers

Relevant parts of the *Project lifting management plan* are presented to the persons involved in lifting operations during on-site induction and specific lifting briefings, including risks involved and preventive measures to be applied. It is followed by an assessment quiz (or similar) to check the level of understanding of the attendees.

This communication is repeated in case of major changes of the *Project lifting management plan* or if deemed necessary, for instance following an accident linked to a lifting operation.

6.2. Lifting team members competencies

Appendix **Competency assessment lifting team members [BYTP-INF-H&S-2136]** sets out the competencies of the various persons involved in lifting operations.

The procedures for conducting regular assessment of the competencies of the various lifting team members are:

xxx,
xxx,
xxx,

[to be completed according to project/site organization]

Documents attesting to their competencies are available on each site during the entire duration of the project.

6.3. Authorization to operate

To operate lifting equipment, one must first obtain an authorization-to-operate issued by the Operator's employer and Project Director/manager.

The Employer and Project Director/Manager establishes and issues the authorization-to-operate based on:

- a medical examination, carried out by a doctor confirming the person is fit to operate the equipment
- a test confirming the operator has the knowledge and know-how to operate the equipment safely
- specific training associated with conditions of use of the equipment and taking the site's constraints into account
- legally required qualifications and certifications

The authorization to operate is established by the project, for activities conducted by CWD, according to the form in **appendix XXX**.

[Insert in appendix the project/site Authorization to Operate form]

7. Equipment conformity

The controls of conformity of the lifting equipment are described in the appendix **Safe Lifting Techniques [BYTP-INF-H&S-2137]** and are mainly relative to the following topics:

- Controls of an equipment before being put in service or being put back in service: Their purpose is to ensure that the equipment meets the prescribed specifications (including those set out in the manufacturer's instruction manual) and that it can be used safely
- Daily inspections of the main controls and main components to confirm in safe condition
- Periodic General Inspections during the utilization of the equipment: Their purpose is to detect damage or deterioration liable to create a hazard. These inspections are carried out by competent and independent persons. They are conducted:
 - at intervals specified by the regulation in force
 - at least every 12 months
 - at least every 6 months for mobile lifting equipment (mobile cranes, forklifts, crane trucks)
- Lifting accessories: visual inspections carried out by the Signallers and Slingers, before starting any lifting operation, with the objective to detect and discard any non-compliant equipment to ensure a safe lifting

Name of persons in charge: **XXX**
XXX
XXX

Role on site: **XXX**
XXX
XXX

8. Lifting management documentation on projects

8.1. Lifting equipment utilization plan

According to the requirements of the *Project lifting management plan*, a *Lifting Equipment Utilization Plan* is prepared by a resource assigned by the project Director/Manager, for each lifting equipment used on the project/site, based on the template **Lifting Equipment Utilization Plan [BYTP-FOR-H&S-2139]**.

Preparation and approval

- a staff member of the technical team on site, a staff member of the Technical Department or a service provider, named in the *Project lifting management plan*, prepares a *Lifting Equipment Utilization Plan* for each lifting equipment used on the site
- the Lifting Supervisor participates in developing the *Lifting Equipment Utilization Plan* for the various lifting equipment used by the teams he supervises on site,
- a resource assigned by the project Director/Manager approves the *Lifting Equipment Utilization Plans* for the various lifting equipment of the site

Responsibilities	NAME	First name	Role
Prepared by			
Approved by			

Content of the *Lifting Equipment Utilization Plan*

For a given lifting equipment, the *Lifting Equipment Utilization Plan* includes:

- description of the lifting equipment
- layout of the lifting equipment
- load charts of the lifting equipment
- human means and qualifications required
- mode of communication (if radio communication is used, operators are equipped with hands-free radios)
- activities and methods of coordination between the various team members
- suitability examination of the lifting equipment, consisting of:
 - formal expression of the needs of the site as to the intended use of the equipment, considering constraints associated with the site where the lifting operation will take place (buried utility lines, overhead power lines, its immediate surroundings, type of soil, bearing capacity, etc.)
 - formal validation of the lifting equipment by the supplier, as to the capacity of this equipment in meeting these needs, under this environment
 - validation of the lifting equipment, upon its arrival on site, regarding its intended use and subsequent change in its location within the site environment throughout its duration of use
 - the siting of the lifting equipment in its various foreseeable working positions (positioning with top and front views and maximum rated capacity according

to the manufacturer's chart), as well as any additional works required to the ground for ensuring its stability (foundations, etc.). This siting factor integrates the elements associated with the dimensions of the lifting equipment, calculations of pressure exerted on the ground by this equipment, and site constraints.

- legally required certifications on site of inspection and test, and load test
- working environment (hazards, obstructions, etc.)
- documents received from the lifting equipment supplier, certifying its conformity with applicable regulations, the follow-up of its periodic general inspections and the carrying out of maintenance prescribed by the manufacturer
- procedures for inspecting lifting equipment and accessories
- analysis of generic risks associated with the use of the lifting equipment
- analysis of risks associated with the specific lifts planned and the environment in which the lifting equipment is used

A copy of the *Lifting Equipment Utilization Plan* is permanently available on each lifting equipment and can be consulted by all persons involved in the operation (for instance for tower cranes: in the operator's cabin and at the bottom of the crane).

8.2. Lifting operation plan

For each lifting operation, for a given lifting equipment and a given load or type of load, the Signaller and/or the lifting Supervisor, in coordination with the project method engineering team, prepares a *Lifting Operation Plan*:

- according to the requirements of the *Project lifting management plan*
- in addition to the *Lifting Equipment Utilization Plan relative to the concerned equipment*
- by using the template **Lifting Operation Plan [BYTP-FOR-H&S-2140]**, contents depend on the type of lifting

During the operation, a copy of the *Lifting Operation Plan* is available at site (location of lifting) and is used as a minimum by the Signaller and the Lifting Operator.

3 types of lifting are defined:

- Simple lifting
- Specific lifting
- Complex lifting

The tool and methodology to help define the type of lift is available within the template **Lifting Operation Plan [BYTP-FOR-H&S-2140]**.

Once the type of lift is determined, the following elements are included in the *Lifting Operation Plan*:

GUIDE TO DEFINE THE TYPE OF LIFT	
Description of the operation :	
Lifting of load not free of the ground or its support (except for specific works requested at project level)	☐ Yes ☐ No
Lifting of load on side pulling force	☐ Yes ☐ No
Lifting of multiple loads or pieces not tied together	☐ Yes ☐ No
Lifting of people (except in case of evacuation with a rescue evacuation cage)	☐ Yes ☐ No
Lifting of load with cable anchoring system	☐ Yes ☐ No
Lifting with uncontrolled worked elements (e.g. balance of calculation, rising, sliding, quality control, load test)	☐ Yes ☐ No
Lifting of construction plant not stopped at the lifting point provided by the supplier or not secured on a suitable lifting platform	☐ Yes ☐ No
Lifting of a heavy suspended load, free to swing, using a wheeled lifting equipment requiring handling and a load indicator	☐ Yes ☐ No
Lifting with a rotation mechanism not driven by an operator specifically trained for such operation	☐ Yes ☐ No
Type of load :	
Lifting of a load free of the ground	☐ Yes ☐ No
Lifting with lifting of a load or a part	☐ Yes ☐ No
Lifting with an extension of a load or a part	☐ Yes ☐ No
Lifting of load with multiple lifting point (scaffold)	☐ Yes ☐ No
In case of any "Yes", the lift is classified as : Complex (see section 8.2.1.1)	
Type of lift :	
Lifting with additional counterweights or a known extension	☐ Yes ☐ No
Lifting in interference with other machinery (gantry, mobile crane, mobile jib)	☐ Yes ☐ No
Lifting with multiple suspension from different points	☐ Yes ☐ No
Lifting with a wheel or roller of a load or a part	☐ Yes ☐ No
Lifting of a load by using the lifting equipment's rotation	☐ Yes ☐ No
Lifting requiring lifting of the load and/or different lengths of string	☐ Yes ☐ No
Lifting with rotation	☐ Yes ☐ No
Lifting of construction plant (except on a lifting point provided by the supplier or on a transport platform designed for operation)	☐ Yes ☐ No
Lifting of a wheeled lifting equipment or a small suspended load, compact, over a short distance, secured to avoid swinging and not able to be worked with the hoist	☐ Yes ☐ No
Use of self-propelled travel device for drivers with specific certification training for such a certified travel device	☐ Yes ☐ No
In case of any "Yes", the lift is classified as : Complex (see section 8.2.1.1)	
Lifting crane in section 8.2.1.1 of the document (see section 8.2.1.1)	
Lifting crane in section 8.2.1.1 of the document (see section 8.2.1.1)	☐ Yes ☐ No
Lifting crane in section 8.2.1.1 of the document (see section 8.2.1.1)	☐ Yes ☐ No
Signatures :	
Signaller	Date
Signaller	Date

Warning : if any "Yes", the operation is not to be carried out. The "Lifting Operation Plan" is valid only if a lifting of the load is planned in the lifting operation. A control is to be carried out before starting the operation.

Lifting operation plan (BYTP-FOR-H&S-2140) 2 pages

"Simple" lifting

If the slinging methods are covered in the document **Standard Slinging [BYTP-INF-H&S-2141]**, only the first page of the *Lifting Operation Plan* is completed, and the operation can be executed.

If the slinging methods are not covered, a *Lifting Operation Plan* shall be developed including the following elements:

- description of the operation
- names of the persons involved in the operation
- the lifting equipment(s) utilized
- the specific load bearing capacity study, if required
- the positioning of the lifting equipment and outriggers, including correctly dimensioned/fabricated pads
- the type of load, its nature, dimensions, weight and packaging
- the methods of slinging, the connection and disconnection of the load (position of the load's center of gravity, lifting points, etc.) – notion of slinging plan
- the type and characteristics of lifting accessories and equipment used (WLL, sizes, etc.)
- the permissible weather conditions (wind, etc.) according to the lifting equipment and load characteristics
- the various requirements and instructions applicable for this lifting operation

"Specific" lifting

In addition to the elements for a "simple" lifting operation, also include:

- overlapping operating zones (e.g. when 2 lifting equipment are installed in such a way that their working radiuses overlap)
- co-activities
- mobilization of additional personnel required, who have the required qualifications and experience
- communication means
- description of the lifting steps (sequencing), with a location on a plan/drawing
- means of barricading and marking to identify the crane's operating zone
- slinging points used or slinging method chosen (choker hitch, basket hitch, etc.)
- calculated angles to be applied + correction coefficients applied
- forces or weights transferred to each leg of lifting accessories
- specific risks analysis conducted by the Lifting Supervisor

"Complex" lifting

In addition to the elements pertaining to a "specific" lifting, establish and implement:

- a Lifting Permit (embedded within the template *Lifting Operation Plan*), issued by the lifting Supervisor

Complex lifting operation cannot commence until the Lifting Permit is approved/signed by the designated approvers, after their checks that all necessary personnel, equipment, technical and safety measures are in place.

Responsibilities	NAME	First name	Role
Lifting Permit approver			

9. Preparation and execution of lifting operations

9.1. Installation of lifting equipment

The lifting equipment are installed and used such a way as to mitigate the risks for their users and for the personnel in their vicinity.

Their positions are marked on the site layout drawing, at least for the regular locations of the various equipment.

It is essential to ensure that the soil bearing capacity is sufficient. If not, it will be necessary to provide larger distribution plates or modify the location of the work equipment

The appendix **Safe Lifting Techniques [BYTP-INF-H&S-2137]** describes, amongst other topics, the measures relative to the installation of lifting equipment according to:

- the presence of aerial electrical lines
- the presence of sensitive buried/underground utilities
- the interferences between 2 or more lifting equipment and other equipment (including concrete trucks, dump trucks, MEWPs)

9.2. Review before start

Before starting any lifting operation, the Signaller for the "simple" lifts and the Lifting Supervisor for the "specific" and "complex" lifts, checks at site that:

- the *Lifting Operation Plan* is prepared and completed
- the *Lifting Equipment Utilization Plan* is prepared and approved
- the *Lifting Permit* is approved (applicable only for "complex" lifting)
- the conditions for carrying out the lifting operation are in accordance with the plans and instructions (technical, personnel and environmental conditions)
- the authorized and required members of the lifting team are present
- the roles and responsibilities are clearly assigned and communicated to each person involved
- the means of communication are available and are operational
- the preventive measures from the generic and specific risk analyses are in place
- the lifting team members are briefed

The Slinger for the "simple" lifts and the Lifting Supervisor for the "specific" and "complex" lifts authorizes the lifting operation to start only if all the above elements are checked, compliant and implemented.

9.3. Lifting execution

Lifting operations are carried out in accordance with the directions set out in **Safe Lifting Techniques [BYTP-INF-H&S-2137]**, covering the following topics:

- operational requirements prior to and during the lift
- slinging methods
- management of lifting accessories

At all times during a lifting operation, the minimum present personnel are:

- a Lifting Operator who must not act as Signaller or Slinger (can be accepted for crane lorry if assessed safe to do so)

- a Signaller who must not interact with the load during its lifting (meaning once the load leaves the ground)
- one or several Slingers if the load needs to be guided

The forbidden practices are as a minimum:

- no lifting of multiple items not tied or bound together securely
- no side-pull lifting or side pulling (except for specifically designed winch operations)
- no lifting of loads that are not free of the ground or its support (except for specialist ground engineering works e.g. pile casing extraction)
- forbidden to disable or bypass/override any safety device

Slinging is conducted according to the slinging practices outlined in the appendix **Standard Slinging [BYTP-NF-H&S-2141]**.

If, for a given lifting operation, a standard lifting arrangement cannot be implemented or if the slinging is not described in the part of the Standard Slinging practices, then a specific slinging method is defined by the Lifting Supervisor and/or by the Methods Engineer and is included in the **Lifting Operation Plan [BYTP-FOR-H&S-2140]** during the preparation of the operation

10. Loan of lifting equipment

In the case of lifting operations carried out by subcontractors using lifting equipment belonging to CWD, the subcontractor and CWD draw up a formalized agreement for the provision and use of such equipment

The subcontractor remains responsible for the lifting operations he carries out, regardless of the origin of the lifting equipment and accessories used under the provision of the equipment loan agreement.